

EDUCATION

Brown University, *Master of Science, Visual Computing & AI*

(Fall 2024 - May 2025)

Brown University, *Bachelor of Science, Computer Science*

(Fall 2020 - May 2024)

PROFESSIONAL EXPERIENCE

C++ Software Engineer

(Jun 2025 - Present)

Dassault Systemes – SIMULIA

- Ship critical fixes and core optimizations to enterprise clients, reducing application latency and improving reliability by **40%**.
- Write **Jasmine** test suites simulating complete E2E user workflows (e.g., login → simulate), increasing UI testing by **80%**.
- Develop **Python** tool for trace analysis to identify performance gaps, automating a previously time-intensive task.
- Piloting team's **Cursor AI** integration and design of internal LLM to assist with error triage and engineering efficiency.

Fullstack Software Engineer Intern,

(May 2024 - Aug 2024)

LI-COR Biosciences – Formerly Onset Data Loggers

- Engineered MQTT handler processing **10k+** IoT messages/day, accelerating device-to-cloud communication by **100%+**.
- Built full-stack **cloud platform** to consolidate and manage firmware files, significantly reducing manual configuration time.
- Prototyped a weather map dashboard with **Leaflet** and **GIS APIs**, enabling real-time user interaction with live weather data.

Head Teaching Assistant

(Oct 2023 - May 2024)

Brown University CS

- Led a **14-person** TA team, managing grading, office hours, and course materials for an upper-level CS course.
- Conducted major overhaul of course website and designed **6** core projects serving **100+ students** per semester.
- Coordinated portfolio reviews with **11** industry professionals to provide practical feedback for students.

Frontend Software Engineer Intern

(May 2023 - Aug 2023)

LI-COR Biosciences – Formerly Onset Data Loggers

- Created a notification system for users to monitor various weather sensors with real-time data visualization.
- Overhauled dashboard and login experience for core environmental-monitoring SaaS platform, supporting **1M+** devices.
- Improved middleware API efficiency, reducing overall latency for end users by **70%**.

Founder & Content Creator

(Jan 2020 - Dec 2023)

Independent Art Business

- Grew a digital art brand to **170k+** followers, gaining millions of views and engagement through A/B testing and analytics.
- Sold **300+** digital prints and custom merchandise generating **\$1.6k profit** in 3 months.
- Secured **4 official brand partnerships** with digital drawing tablet companies such as Huion and XPPen.

PROJECTS

Origin Two / Ninja Mouse (2D + 3D Game Engines)

- Built **Java** and **C++ game engines** with a reusable UI and sound toolkit, animation cycling, and physics and collision systems.
- Shipped **4 games** with additional procedural level loading and NPC AI.

Realtime Ocean Water Simulation

- Implemented a physically accurate ocean water simulation based on motion-picture oceans.
- Built a **fast fourier transform optimizer** to allow real-time user manipulation of wind speed and turbulence.
- Introduced new features ocean foam, particle systems for spray, caustics, and custom shaders for reflection and refraction.

Realtime Graphics & Ray Tracing

- Created a **real-time C++/OpenGL** scene viewer with shape and camera manipulation and custom GLSL shaders.
- Integrated ray tracing algorithm for realistic rendering using the Phong model, mirror reflections, and texture mapping.
- Programmed custom toon and outline shaders with framebuffer objects to achieve stylized post-processing effects.

3D Modeling & Animation

- Animated game-ready and cyclable attack sequences with a custom 3D rig and character in Unreal engine.
- Designed, modeled, rigged, textured, and animated **2** original characters with custom shaders for an animated short.
- Created adaptable shader and texture customization sliders, supporting rapid creation of varying environments.

SKILLS

Computer Languages: C++, Python, Java, OpenGL, GLSL, TensorFlow, React.js, Vue.js, Node.js, JS, TS, Shell, HTML/CSS

Tools: Unreal Engine 5, Maya, Blender, Gaea, Adobe, Git, Docker, Figma, Kanban, Atlassian/Jira, Agile Methodologies